

# Fairness and Proxy Bias in Occupation Prediction: Scaling, Masking, and Token-level Attribution across Model Families

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## Abstract

We study gender bias in occupation prediction using the Bias-in-Bios dataset, comparing Pythia generative models (160M–1.4B) and encoder classifiers (DistilBERT, RoBERTa) under masked and unmasked conditions. Against the common intuition that larger models are fairer, we find that unsupervised scaling *amplifies* bias in the Pythia family on this task. Gender masking may reduce fairness gaps in encoder models, though the effect is small and close to sampling variability; it produces backfire effects in generative models, indicating that proxy representations may be model-specific rather than a universal property of the input text. Token-level attribution further suggests that indirect contextual signals, not solely explicit gender markers, appear to be the dominant carriers of the gender signal — and that they persist even after masking.

## 1 Introduction

Language models encode and reproduce societal biases present in their training data [Caliskan et al., 2017]. When deployed in high-stakes settings, such biases can translate into systematic disparities that disadvantage particular demographic groups.

Occupation prediction is a revealing case study for this problem. Models trained on real-world career data tend to associate professions like *nurse* or *dietitian* with female-typical language, and *surgeon* or *software engineer* with male-typical patterns. De-Arteaga et al. [2019] showed, using the Bias-in-Bios dataset, that scrubbing explicit gender signals does not remove these disparities: implicit proxy features survive. Yet the identity of these proxy features — which tokens are associated with the gender signal, how their salience changes under masking, and whether this varies with model scale or model type — remains poorly understood.

## Contributions.

- **Proxy bias is not solely driven by explicit gender markers.** On the Bias-in-Bios task, gender-correlated contextual tokens dominate attribution scores by a large margin over explicit gender words, while remaining active even after gender masking. To our knowledge, this provides one of the first detailed token-level attribution analyses of this kind on this dataset.
- **Masking effectiveness depends on training regime, not architecture alone.** Explicit gender masking reduces bias in encoder models and in zero/few-shot generative settings, but proves ineffective—and sometimes harmful—after generative fine-tuning. Attribution stability further shows that proxy representations differ between individual models (e.g. DistilBERT vs. RoBERTa) even between closely related encoder models, providing evidence that proxy structure may be model-specific rather than a fixed property of the input text.
- **Unsupervised scaling amplifies bias; supervision recovers it.** Across Pythia model sizes (160M–1.4B), zero-shot bias grows substantially with scale, while fine-tuning consistently restores fairness. Both findings hold across masked and unmasked conditions on this dataset, this suggests the pattern of capacity-driven bias amplification may extend to other settings.

## 2 Related Work

**Bias in occupation prediction.** De-Arteaga et al. [2019] introduced Bias-in-Bios and showed that classifiers exhibit gender disparities even after pronoun scrubbing. Romanov et al. [2019] examined the limits of debiasing through scrubbing. Our

work extends this line of research by identifying and quantifying the specific proxy tokens responsible, using gradient-based attribution.

**Fairness metrics.** [Hardt et al. \[2016\]](#) formalised Equalized Odds and [Dwork et al. \[2012\]](#) introduced demographic parity. We evaluate both using a one-vs-rest multiclass extension macro-averaged over occupations.

**Attribution methods.** [Sundararajan et al. \[2017\]](#) introduced Integrated Gradients (IG), which satisfies completeness and sensitivity axioms and is the standard choice for token-level analysis of sequence classifiers [[Attanasio et al., 2022](#)]; its application to occupation classification is limited.

**Model scaling and fairness.** [Bommasani et al. \[2021\]](#) discusses bias amplification as a risk in large foundation models. Our scaling study gives concrete evidence of this in the Pythia family: zero-shot bias increases with model size.

**Gender masking.** [Lu et al. \[2020\]](#) study counterfactual data augmentation for gender bias. Unlike that work, we also measure how masking affects token-level attribution patterns, giving an internal view of how models redistribute feature importance when gender signals are removed.

**Debiasing methods.** Proposed approaches include adversarial training [[Zhao et al., 2018](#)], data augmentation [[Lu et al., 2020](#)], and post-hoc embedding projection [[Sun et al., 2019](#)]. [Gonen and Goldberg \[2019\]](#) showed such methods often conceal rather than remove bias, motivating diagnostic analysis of which features carry residual signals.

### 3 Dataset and Preprocessing

#### 3.1 Bias-in-Bios

We use the Bias-in-Bios dataset [[De-Arteaga et al., 2019](#)] (LabHC/bias\_in\_bios), a standard benchmark for studying occupational stereotypes and evaluating debiasing methods in NLP. Each example consists of a short biography (`hard_text`) paired with an integer profession label and a binary gender attribute inferred from pronoun usage by the original authors ( $0 \rightarrow \text{M}$ ,  $1 \rightarrow \text{F}$ ). For example: “*Dr. K. Rama Krishna practices at Pet Park Clinic in Begumpet, Hyderabad. He completed B.V.Sc A.H from College of Veterinary Science, Rajendranagar, SVVU in 2009.*” (label: *physician*, gender: M). In practice, models may produce biased outputs on similar inputs. For instance, a biography describing caregiving activities may be predicted as *nurse* when written with female-

associated language, but as *physician* when similar qualifications are expressed with male-associated language. This illustrates how learned associations, rather than qualifications alone, can influence predictions. The dataset contains 381,807 examples (train 248,141 / dev 38,198 / test 95,468), with a near-identical gender split of  $\approx 46.2\%$  female across all partitions, indicating stratified sampling. We restrict to the **top 20 most frequent occupations** in the training split. The task is to predict occupation from biography text alone; gender is used only for post-hoc fairness evaluation.

Table 1 shows the per-profession counts and gender distribution in the training split. Several occupations exhibit strong gender skew: *dietitian* (92.9% F), *nurse* (90.8% F), and *model* (82.7% F) are heavily female-dominated, while *surgeon* (14.8% F), *software engineer* (15.8% F), and *composer* (16.4% F) are heavily male-dominated.

| Profession        | N (train) | F%   |
|-------------------|-----------|------|
| professor         | 76,748    | 45.1 |
| physician         | 26,648    | 49.4 |
| attorney          | 21,169    | 38.3 |
| photographer      | 15,773    | 35.7 |
| journalist        | 12,960    | 49.5 |
| nurse             | 12,316    | 90.8 |
| psychologist      | 11,945    | 62.1 |
| teacher           | 10,531    | 60.2 |
| dentist           | 9,479     | 35.3 |
| surgeon           | 8,829     | 14.8 |
| architect         | 6,568     | 23.7 |
| painter           | 5,025     | 45.7 |
| model             | 4,867     | 82.7 |
| filmmaker         | 4,545     | 32.9 |
| poet              | 4,558     | 49.0 |
| software engineer | 4,492     | 15.8 |
| composer          | 3,637     | 16.4 |
| accountant        | 3,660     | 36.7 |
| dietitian         | 2,567     | 92.9 |
| comedian          | 1,824     | 21.1 |

Table 1: Per-profession training counts and F% ( $\text{M}\% = 100 - \text{F}\%$ ). Distribution is consistent across all splits.

#### 3.2 Evaluation Subset

All models are evaluated on a **3,000-sample test subset**, stratified proportionally by profession from the 95,468-example test split; IG cost (32 forward passes per example) precluded the full test set. Per-profession counts range from 932 (professor) to 27 (comedian), so results for rare occupations should

be read with caution.

### 3.3 Gender Masking

We evaluate all models under two conditions: **Unmasked** (original text) and **Masked**, where gender tokens are replaced with the model’s native mask token ([MASK] for DistilBERT, <mask> for RoBERTa, and the literal string [MASK] for Pythia, which as a causal language model has no native mask token) using a case-insensitive word-boundary regex. Three categories of tokens are masked: pronouns and possessives (*he, she, his, her, etc.*), titles and honorifics (*mr, mrs, ms, sir, lady, etc.*), and 54 gendered nouns (*father, mother, husband, wife, actor, actress, etc.*). Names are not masked, as they carry biographical content unrelated to gender and reliable name-matching produces too many false positives.

## 4 Methods

### 4.1 Pythia: Zero-shot and Few-shot

We evaluate Pythia [Biderman et al., 2023] at three scales (160M, 410M, and 1.4B parameters) under zero-shot (biography plus candidate label list) and few-shot ( $K = 4$  exemplar pairs) regimes, both with and without gender masking.<sup>1</sup> As a causal language model, Pythia scores each candidate label  $c_k$  by its joint log-probability given the prompt  $p$ :

$$\log P(c_k | p) = \sum_{t=1}^{|c_k|} \log P\left(c_k^{(t)} | p, c_k^{(1)}, \dots, c_k^{(t-1)}\right) \quad (1)$$

The predicted label is  $\hat{y} = \arg \max_{c_k} \log P(c_k | p)$ .

### 4.2 Pythia: Supervised Fine-tuning

Pythia is fine-tuned as a sequence classifier (with a linear head over the final hidden state) using LoRA [Hu et al., 2022] on the attention projections ( $r = 16$ ,  $\alpha = 32$ , dropout 0.1). We train for 3 epochs with a batch size of 16 and a max length of 256. The learning rate is  $3 \times 10^{-4}$  for Pythia 160M and  $2 \times 10^{-5}$  otherwise. Checkpoints are selected by the development set Macro-F1. For the 1.4B variant, 4-bit QLoRA is utilized.

<sup>1</sup>The four exemplars cover professor (female pronoun), physician (male pronoun), journalist (male pronoun), and attorney (female pronoun), giving a balanced 2:2 female-to-male ratio.

### 4.3 Encoder Baselines: DistilBERT and RoBERTa

DistilBERT [Sanh et al., 2019] (66M) and RoBERTa-base [Liu et al., 2019] (125M) are fine-tuned as sequence classifiers with a linear head, lr  $2 \times 10^{-5}$ , batch 16, 3 epochs, max length 256, best checkpoint by Macro-F1. The same pipeline runs for both masked and unmasked conditions with a shared label mapping.

### 4.4 Proxy Bias Attribution

We apply Integrated Gradients [Sundararajan et al., 2017] to the two fine-tuned encoder models, computing attributions toward the *predicted* class index. We use **32 interpolation steps** and a **zero-embedding baseline**. Token-level scores are obtained by summing attribution values across the embedding dimension for each token position. Before top- $K$  selection, subword tokens are merged to word level: BERT-style ## continuation pieces and RoBERTa’s whitespace-prefixed BPE tokens are each handled separately, with scores summed across pieces. We keep only tokens with positive attribution scores and a minimum length of 2 characters, with a maximum sequence length of **256 tokens**.

The pipeline runs in two stages. In **Stage 1** ( $K_1 = 5$ ), the 5 highest-scoring positive tokens are extracted per example. In **Stage 2**, these are aggregated across all examples for each profession (minimum token length 3, stopwords removed) to compute a weighted score:

$$\text{wscore}(t, p) = \text{count\_topk}(t, p) \times \overline{\text{attr}}(t, p) \quad (2)$$

where  $\text{count\_topk}(t, p)$  is the number of examples where token  $t$  appears in the stage-1 top-5 for profession  $p$ , and  $\overline{\text{attr}}(t, p)$  is the mean attribution score. We attribute toward the *predicted* label rather than the true label throughout, since the goal is to understand what the model relies on, not whether it is correct [Sundararajan et al., 2017]. Note that gendered pronouns (*he, she, his, her, etc.*) are included in the Stage 2 stopword list and therefore never appear in the aggregated proxy tables, regardless of masking condition. The gender signal comparison in Table 6 thus reflects only non-pronoun gendered vocabulary (titles, gendered nouns). Faithfulness testing is unaffected, as it uses the Stage 1 top- $k$  tokens which retain pronouns.

**Stability analysis.** To compare masked and unmasked attribution patterns, we take the **top-10**

**aggregated tokens** per profession ( $K_2 = 10$ ) and compute a shared token ratio:

$$r(p) = \frac{|\mathcal{T}_{\text{sh}}|}{|\mathcal{T}_{\text{sh}}| + |\mathcal{T}_{\text{m}}| + |\mathcal{T}_{\text{u}}|} \quad (3)$$

where  $\mathcal{T}_{\text{sh}}$ ,  $\mathcal{T}_{\text{m}}$ ,  $\mathcal{T}_{\text{u}}$  are the tokens shared between conditions, masked-only, and unmasked-only respectively. Gender-conditioned proxy tables are produced using the same procedure, stratified by profession and gender.

**Faithfulness validation.** To check whether attributed tokens are load-bearing, we replace the top-5 in each example’s text with [MASK] and re-run the model, measuring the probability drop:

$$\text{Faith}(x) = P(\hat{y} | x) - P(\hat{y} | x_{\text{erased}}) \quad (4)$$

where  $\hat{y}$  is the predicted label for example  $x$ . Attribution scores are correlational: high attribution indicates token salience under the learned representation, not causal responsibility. Erasure provides limited interventional support but is best read as local functional dependence rather than proof of mechanism.

## 5 Experimental Setup

Code and prediction files are available at <https://github.com/usukhbayarp/comp0087-fairness-occupation-prediction>.

| Model       | Type | Regime                  | Cond. |
|-------------|------|-------------------------|-------|
| Pythia-160M | Gen. | zero-shot, few-shot, FT | M, U  |
| Pythia-410M | Gen. | zero-shot, few-shot, FT | M, U  |
| Pythia-1.4B | Gen. | zero-shot, few-shot, FT | M, U  |
| DistilBERT  | Enc. | Fine-tuned              | M, U  |
| RoBERTa     | Enc. | Fine-tuned              | M, U  |

Table 2: All model configurations. FT=fine-tuned; M=masked; U=unmasked.

### 5.1 Evaluation Metrics

We evaluate on both task performance and fairness. Let  $Y$  be the true occupation label,  $\hat{Y}$  the predicted label,  $A \in \{M, F\}$  the binary gender attribute, and  $\mathcal{C}$  the set of 20 occupation classes.

**Task performance.** We report Accuracy and Macro-F1 to account for class imbalance.

**Fairness metrics.** We measure gender disparities using one-vs-rest multiclass extensions of Demographic Parity (DP) and Equalized Odds (EO), macro-averaged across occupations [Hardt et al.,

2016, Dwork et al., 2012]. For binary gender  $A \in \{M, F\}$  and label set  $\mathcal{C}$ :

$$\Delta\text{DP} = \frac{1}{|\mathcal{C}|} \sum_c |P(\hat{Y}=c | A=M) - P(\hat{Y}=c | A=F)|$$

$$\Delta\text{TPR} = \frac{1}{|\mathcal{C}|} \sum_c |P(\hat{Y}=c | Y=c, A=M) - P(\hat{Y}=c | Y=c, A=F)|$$

$$\Delta\text{FPR} = \frac{1}{|\mathcal{C}|} \sum_c |P(\hat{Y}=c | Y \neq c, A=M) - P(\hat{Y}=c | Y \neq c, A=F)|$$

The EO gap is  $\max(\Delta\text{TPR}, \Delta\text{FPR})$ .

## 6 Results and Discussion

Results are based on the 3,000-sample evaluation subset; some variance is expected due to subset size.

### 6.1 Task Performance

| Model                 | C. | Acc.  | F1    |
|-----------------------|----|-------|-------|
| Pythia-160M FT        | U  | 0.390 | 0.125 |
| Pythia-160M FT        | M  | 0.382 | 0.094 |
| Pythia-410M FT        | U  | 0.581 | 0.323 |
| Pythia-410M FT        | M  | 0.576 | 0.325 |
| Pythia-1.4B FT        | U  | 0.877 | 0.848 |
| Pythia-1.4B FT        | M  | 0.877 | 0.852 |
| DistilBERT FT         | U  | 0.875 | 0.843 |
| DistilBERT FT         | M  | 0.872 | 0.844 |
| RoBERTa FT            | U  | 0.882 | 0.851 |
| RoBERTa FT            | M  | 0.880 | 0.851 |
| Pythia-160M zero-shot | U  | 0.402 | 0.227 |
| Pythia-160M zero-shot | M  | 0.370 | 0.208 |
| Pythia-160M few-shot  | U  | 0.353 | 0.243 |
| Pythia-160M few-shot  | M  | 0.334 | 0.244 |
| Pythia-410M zero-shot | U  | 0.531 | 0.554 |
| Pythia-410M zero-shot | M  | 0.530 | 0.562 |
| Pythia-410M few-shot  | U  | 0.364 | 0.378 |
| Pythia-410M few-shot  | M  | 0.354 | 0.360 |
| Pythia-1.4B zero-shot | U  | 0.658 | 0.567 |
| Pythia-1.4B zero-shot | M  | 0.618 | 0.518 |
| Pythia-1.4B few-shot  | U  | 0.512 | 0.518 |
| Pythia-1.4B few-shot  | M  | 0.489 | 0.491 |

Table 3: Task performance results. U=unmasked; M=masked.

Pythia-1.4B fine-tuned reaches Accuracy = 0.877, close to the encoder baselines (RoBERTa 0.880, DistilBERT 0.872). Smaller fine-tuned Pythia models underperform their zero-shot counterparts (e.g., 410M F1: 0.554→0.323), suggesting



insufficient capacity for stable LoRA fine-tuning. Zero-shot F1 improves with scale (0.23→0.57, 160M→1.4B); few-shot degrades at larger scales (410M: 0.55→0.38; 1.4B: 0.57→0.51), suggesting difficulty leveraging in-context examples. Masking has little effect on task performance in the fine-tuned regime.

## 6.2 Fairness Results

| Model                 | C.       | EO           | DP           | TPR          | FPR          |
|-----------------------|----------|--------------|--------------|--------------|--------------|
| <b>RoBERTa FT</b>     | <b>U</b> | <b>0.082</b> | <b>0.024</b> | <b>0.082</b> | <b>0.005</b> |
| RoBERTa FT            | M        | 0.063        | 0.022        | 0.063        | 0.004        |
| <b>DistilBERT FT</b>  | <b>U</b> | <b>0.088</b> | <b>0.022</b> | <b>0.088</b> | <b>0.004</b> |
| DistilBERT FT         | M        | 0.071        | 0.022        | 0.071        | 0.004        |
| <b>Pythia-1.4B FT</b> | <b>U</b> | <b>0.086</b> | <b>0.023</b> | <b>0.086</b> | <b>0.004</b> |
| Pythia-1.4B FT        | M        | 0.080        | 0.022        | 0.080        | 0.003        |
| <b>Pythia-1.4B FS</b> | <b>U</b> | <b>0.120</b> | <b>0.025</b> | <b>0.120</b> | <b>0.011</b> |
| Pythia-1.4B FS        | M        | 0.083        | 0.017        | 0.083        | 0.008        |
| <b>Pythia-1.4B 0s</b> | <b>U</b> | <b>0.209</b> | <b>0.029</b> | <b>0.209</b> | <b>0.019</b> |
| Pythia-1.4B 0s        | M        | 0.134        | 0.014        | 0.134        | 0.005        |
| <b>Pythia-410m FT</b> | <b>U</b> | <b>0.085</b> | <b>0.012</b> | <b>0.085</b> | <b>0.006</b> |
| Pythia-410m FT        | M        | 0.088        | 0.011        | 0.088        | 0.006        |
| <b>Pythia-410m FS</b> | <b>U</b> | <b>0.108</b> | <b>0.019</b> | <b>0.108</b> | <b>0.010</b> |
| Pythia-410m FS        | M        | 0.103        | 0.014        | 0.103        | 0.008        |
| <b>Pythia-410m 0s</b> | <b>U</b> | <b>0.135</b> | <b>0.038</b> | <b>0.135</b> | <b>0.021</b> |
| Pythia-410m 0s        | M        | 0.095        | 0.024        | 0.095        | 0.012        |
| <b>Pythia-160m FT</b> | <b>U</b> | <b>0.042</b> | <b>0.012</b> | <b>0.042</b> | <b>0.008</b> |
| Pythia-160m FT        | M        | 0.027        | 0.005        | 0.027        | 0.004        |
| <b>Pythia-160m FS</b> | <b>U</b> | <b>0.065</b> | <b>0.012</b> | <b>0.065</b> | <b>0.009</b> |
| Pythia-160m FS        | M        | 0.069        | 0.009        | 0.069        | 0.006        |
| <b>Pythia-160m 0s</b> | <b>U</b> | <b>0.085</b> | <b>0.027</b> | <b>0.085</b> | <b>0.025</b> |
| Pythia-160m 0s        | M        | 0.058        | 0.012        | 0.058        | 0.009        |

Table 4: Fairness metrics per model configuration. EO = Equalized Odds gap; DP = Demographic Parity gap; TPR = True Positive Rate gap; FPR = False Positive Rate gap; M/U = masked/unmasked; FT/FS/0s = fine-tuned/few-shot/zero-shot; EO = max{FPR, TPR}

Fine-tuned models consistently show lower EO gaps than their zero-shot counterparts. Unmasked zero-shot EO grows from ~0.09 at 160M to ~0.21 at 1.4B, showing that unsupervised scaling amplifies bias in the Pythia family.

Crucially, the effectiveness of gender masking depends heavily on the learning regime. We observe a large masking benefit for Pythia in zero-shot and few-shot settings (e.g., Pythia-1.4B 0s: 0.209→0.134; Pythia-410M 0s: 0.135→0.095; Pythia-1.4B FS: 0.120→0.083), though zero-shot bias still grows with scale even when masked.

In contrast, masking produces a comparatively smaller difference for the fine-tuned encoder models (RoBERTa: 0.082 to 0.063; DistilBERT: 0.088

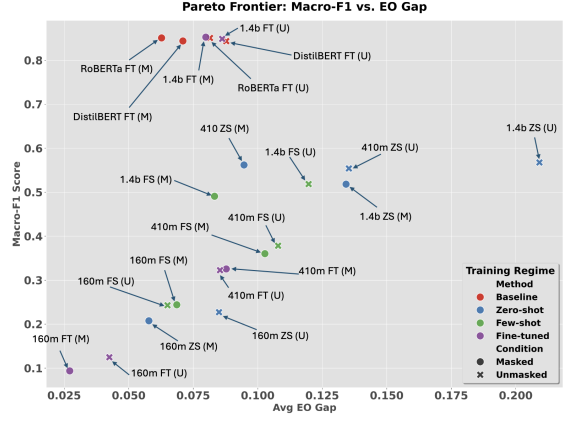


Figure 1: Pareto frontier of Macro-F1 vs. Equalized Odds gap. RoBERTa masked fine-tuned, followed by DistilBERT, is the most Pareto-optimal model. The generative models which are masked and with largest capacity, as well as the baseline encoders, strongly dominate lower-capacity, zero-shot and few-shot alternatives.

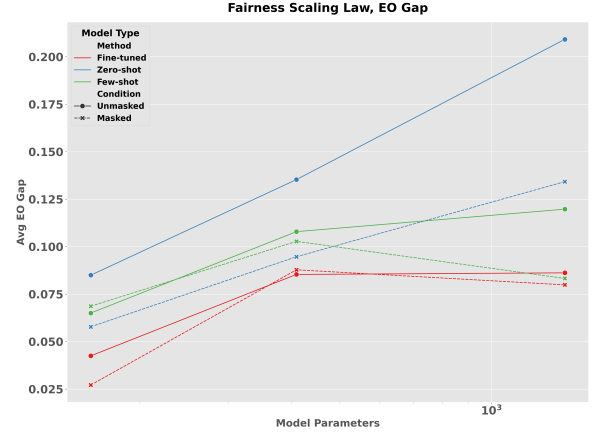


Figure 2: Scaling behavior of Equalized Odds gap across model sizes, 1.4B (red), 410m (green), and 160m (blue). The solid line corresponds to unmasked and dashed to masked. Zero-shot bias increases with scale, while fine-tuning mitigates this effect.

to 0.071). Furthermore, at the fine-tuned level for Pythia, masking produces backfire effects at the profession level (e.g., for *teacher*, *poet*, and *painter*), with Pythia-410M FT showing a small aggregate increase in bias (0.085 to 0.088).

Finally, the FPR gap is consistently much smaller than the TPR gap across all models, so TPR disparity remains the dominant fairness concern. Taken together, masking successfully disrupts bias in raw pre-trained representations (zero/few-shot), but remains insufficient once a model is explicitly fine-tuned. Per-occupation fairness metrics for all professions and models are shown in Appendix C.

### 6.3 Proxy Bias Analysis

We examine which tokens drive occupation predictions and how their importance shifts under masking, via attribution stability (§6.3), gender signal identification (§6.3), erasure faithfulness (§6.3), and counterfactual pronoun swap (§6.3).

#### Step 1: Attribution Stability Under Masking

| Profession        | DistilBERT | RoBERTa |
|-------------------|------------|---------|
| accountant        | 0.43       | 0.82    |
| architect         | 0.43       | 0.82    |
| attorney          | 0.67       | 0.82    |
| comedian          | 0.54       | 0.33    |
| composer          | 0.54       | 0.43    |
| dentist           | 0.54       | 0.33    |
| dietitian         | 0.67       | 0.54    |
| filmmaker         | 0.67       | 0.54    |
| journalist        | 0.67       | 0.67    |
| model             | 0.82       | 0.43    |
| nurse             | 0.43       | 0.54    |
| painter           | 0.67       | 0.54    |
| photographer      | 0.67       | 0.67    |
| physician         | 0.67       | 0.54    |
| poet              | 0.54       | 0.67    |
| professor         | 0.82       | 1.00    |
| psychologist      | 0.82       | 0.82    |
| software engineer | 0.67       | 0.43    |
| surgeon           | 0.67       | 0.67    |
| teacher           | 0.82       | 0.82    |

Table 5: Shared token ratio (masked vs. unmasked) for all 20 professions. Higher = more stable. Ratio = shared / (shared + masked-only + unmasked-only) over the top-10 aggregated tokens.

Under RoBERTa, *comedian* and *dentist* are the most unstable professions (ratio 0.33), yet under DistilBERT both are moderately stable (0.54). The same reversal appears for *nurse* (DistilBERT 0.43, RoBERTa 0.54). Under DistilBERT, *accountant* and *architect* reach the same minimum ratio as *nurse* (0.43), while both are substantially more stable under RoBERTa (0.82), showing that instability is not specific to gender-skewed professions. *Professor* achieves maximum stability under RoBERTa (1.00), with identical top-10 tokens across masked and unmasked conditions. These cross-profession reversals suggest that proxy stability is model-specific, not simply a function of training data.

Label leakage is expected (profession names frequently appear in top- $K$ ); the informative signal lies in secondary token shifts. Under Distil-

BERT, *nurse: practitioner* drops 26.51→11.18 and *graduated* drops 19.11→0 (disappears entirely) after masking, while *especially* (15.52) gains weight. Under RoBERTa, *physician: medicine* rises 166.64→292.25 and *patients* gains prominence (24.34→52.72), entering the top-5. Detailed per-profession token distributions are provided in Appendix A.

#### Step 2: Gender Signal Identification

| Model      | Token           | Type       | W.Score |
|------------|-----------------|------------|---------|
| DistilBERT | <i>actor</i>    | gender     | 1.15    |
| DistilBERT | <i>sister</i>   | gender     | 1.13    |
| DistilBERT | <i>mrs</i>      | gender     | 1.05    |
| RoBERTa    | <i>mrs</i>      | gender     | 2.11    |
| RoBERTa    | <i>sister</i>   | gender     | 1.13    |
| DistilBERT | <i>nursing</i>  | profession | 55.53   |
| DistilBERT | <i>medicine</i> | profession | 268.20  |
| DistilBERT | <i>research</i> | profession | 329.52  |
| RoBERTa    | <i>nursing</i>  | profession | 70.44   |
| RoBERTa    | <i>medicine</i> | profession | 166.64  |
| RoBERTa    | <i>research</i> | profession | 333.32  |

Table 6: Attribution weight of gender tokens vs. dominant profession tokens, within each model. Gender tokens appear in both models but with substantially smaller weights than profession tokens. These aggregated scores are derived from the per-profession token distributions reported in Appendix A.

Both models show a small number of explicit gender tokens in the aggregated top- $K$  features, but their weights are much lower than those of dominant profession tokens. In DistilBERT, the highest-ranked gender token (*actor*: 1.15) is  $48\text{--}287\times$  smaller than the top profession tokens (55.53–329.52). In RoBERTa, the highest gender token (*mrs*: 2.11) is  $33\text{--}158\times$  smaller (profession range: 70.44–333.32). This suggests that gender-correlated information appears to be encoded primarily through indirect contextual signals rather than explicit gender references, consistent with models exploiting statistical co-occurrences in the training corpus [De-Arteaga et al., 2019, Gonen and Goldberg, 2019]: gendered role vocabulary, credential terminology, and contextual framing appear to encode gender-correlated information without any explicit reference to gender.

Gender-conditioned analysis shows partial asymmetry. Under DistilBERT, *nurse|F* and *nurse|M* both retain 5 shared tokens (ratio 0.33), so there is no directional difference between genders. Under

RoBERTa, nurse|F retains 7 shared tokens (ratio 0.54) versus nurse|M at 6 (ratio 0.43), a modest asymmetry suggesting the model’s feature space for *nurse* is slightly more anchored in female-typical contextual patterns. This aligns with the fairness results in Section 6.2, where *nurse* shows one of the largest EO gap reductions under masking, suggesting that professions with less stable proxy representations benefit more from masking.

### Qualitative Illustrations

**Example 1: Gender asymmetry in attributed tokens (nurse, unmasked).** Both biographies are correctly predicted as *nurse*.

| Male biography excerpt                                                                                                                                                               | Female biography excerpt                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “He holds a Masters of Nursing Degree and has served in the Australian Defense force in various roles for over 32 years. He has <b>presented</b> in this field for over 20 years...” | “Her <b>scope</b> of practice encompasses <b>health</b> promotion, disease prevention, diagnosis and management of common and complex <b>health</b> care problems... throughout the <b>aging</b> process...” |
| Top tokens: <i>nursing</i> (3.18), <i>masters</i> (0.61), <i>presented</i> (0.59), <i>australian</i> (0.44)                                                                          | Top tokens: <i>scope</i> (0.74), <i>health</i> (0.66), <i>aging</i> (0.61), <i>practice</i> (0.55), <i>her</i> (0.49)                                                                                        |

The male prediction anchors on explicit credentials (*nursing*, *masters*), while the female prediction distributes across social-care context words (*scope*, *aging*), consistent with the nurse|M faithfulness asymmetry (0.756 vs. 0.354).

**Example 2: Prediction flip after masking (backfire).**

“[MASK] is a parent of a child with autism, and is dedicated to providing professional assistance to other parents... [MASK] can be reached at (408) 843-7997, and is available for paid phone consultation...”

True label: *psychologist*. Unmasked prediction: *psychologist* ✓ — top tokens: *parent* (1.15), *autism* (0.81), *child* (0.39), *professional* (0.33). Masked prediction: *attorney* × — top tokens: *408* (0.63), *autism* (0.34), *reached* (0.33), *parents* (0.28).

After masking *He*, the parent/child anchor loses salience and the phone number and “consultation” framing pattern-match *attorney*, producing an error rather than a fairness improvement.

**Example 3 (mini): Indirect proxy tokens driving a physician prediction.**

“Ms. Croteau **practices medicine** in New Britain, CT and specializes in Allergy & Immunology. Ms. Croteau is affiliated with **Hospital** of Central Connecticut.”

Predicted: *physician* ✓ — top tokens: *medicine* (1.13), *practices* (0.96), *hospital* (0.52). Clinical role words dominate the attributed tokens; no gender tokens appear in the top 5, consistent with the view that indirect contextual signals, not solely explicit gender markers, are most strongly associated with the predictive signal.

### Step 3: Faithfulness Validation

Integrated Gradients gives correlational evidence of feature importance. The top-*K* erasure test complements this with a direct intervention: we erase the attributed tokens and measure the confidence drop to check whether they are genuinely load-bearing. Table 7 reports mean faithfulness by profession across both models and conditions. Across all 3,000 examples in the unmasked condition, 89.6% (DistilBERT) and 82.3% (RoBERTa) of predictions show positive faithfulness, consistent with the attributed tokens being functional.

| Profession        | DistilBERT |       | RoBERTa |       |
|-------------------|------------|-------|---------|-------|
|                   | U          | M     | U       | M     |
| Dietitian         | 0.620      | 0.743 | 0.548   | 0.372 |
| Nurse             | 0.395      | 0.518 | 0.203   | 0.197 |
| Psychologist      | 0.356      | 0.378 | 0.232   | 0.252 |
| Painter           | 0.473      | 0.418 | 0.339   | 0.342 |
| Physician         | 0.233      | 0.191 | 0.235   | 0.200 |
| Attorney          | 0.162      | 0.221 | 0.175   | 0.158 |
| Professor         | 0.160      | 0.214 | 0.162   | 0.182 |
| Software Engineer | 0.171      | 0.204 | 0.193   | 0.199 |
| Overall           | 0.330      | 0.364 | 0.243   | 0.249 |

Table 7: Mean faithfulness (probability drop after erasing top-5 attributed tokens) for a selected subset of professions. U=unmasked; M=masked. Higher = more functionally important tokens under erasure. The *Overall* row is the macro-average across all 20 professions (equal weight per profession, not per example). Full results for all 20 professions are provided in Appendix D.

Faithfulness scores are broadly consistent with the stability analysis. Professions with the highest faithfulness (*dietitian*: 0.620, *nurse*: 0.395) tend to have lower stability ratios than the most stable professions (*professor*: ratio 0.82, faithfulness 0.160), suggesting that unstable proxy representations co-occur with more concentrated, load-bearing token sets. This supports the view that attribution instabil-

ity identifies professions whose predictions depend on a small, critical token set.

Gender-conditioned faithfulness shows a clear asymmetry for *nurse*: DistilBERT  $\text{nurse}|M = 0.756$  vs.  $\text{nurse}|F = 0.354$ ; RoBERTa  $\text{nurse}|M = 0.533$  vs.  $\text{nurse}|F = 0.163$ , indicating that male nurse representations are concentrated in a smaller critical token set. Full gender-conditioned faithfulness results are provided in Appendix E; a second notable asymmetry appears for *poet* under DistilBERT unmasked (0.434 vs. 0.200).

Masking increases overall faithfulness for both models (DistilBERT: +0.034; RoBERTa: +0.006), suggesting the remaining tokens absorb more predictive load once gender signals are removed.

#### Step 4: Counterfactual Pronoun Swap

To provide limited interventional evidence, we counterfactually swap all gendered pronouns in each biography ( $M \rightarrow F$  or  $F \rightarrow M$ ) and measure whether the predicted occupation changes. Overall flip rates are low: 3.5% for DistilBERT and 3.9% for RoBERTa (mean confidence drop: 0.009 and 0.011 respectively), confirming that explicit gender tokens are not the primary drivers of predictions and reinforcing the proxy-based explanation. However, the consistent drop in prediction confidence suggests that gender information still influences internal representations even when predictions remain unchanged. Profession-level sensitivity varies: *nurse* (16.3% / 13.1%), *surgeon* (13.3% / 13.0%), and *comedian* (11.1% / 16.0%) show the highest flip rates, while *composer* and *dentist* show zero flips across both models; *filmmaker* shows zero flips for DistilBERT but 5.6% for RoBERTa. Masked conditions are excluded: masking replaces all pronouns with [MASK] prior to inference, leaving no tokens to swap.

## 7 Conclusion

Fine-tuned models outperform zero/few-shot baselines on both performance and fairness. Scaling without supervision amplifies bias: zero-shot EO gap grows from  $\sim 0.09$  at 160M to  $\sim 0.21$  at 1.4B, showing that model capacity alone is not a substitute for fine-tuning or explicit debiasing. Masking reduces aggregate EO in encoder models (by 0.017–0.019), although this effect is close to the estimated sampling variability ( $\approx 0.018$ ) and should be interpreted as a weak but consistent trend rather than a robust improvement; masking produces backfire

effects at the profession level in Pythia. Attribution analysis suggests that gender-correlated information appears to be encoded primarily through indirect contextual tokens; their salience is model-specific and their functional importance is supported by erasure faithfulness testing and counterfactual pronoun swap (overall flip rate: 3.5–3.9%). Gender-conditioned analysis adds a further observation: male nurse predictions are concentrated in a critically small feature set, with faithfulness nearly double that of female nurse predictions (0.756 vs. 0.354 under DistilBERT).

**Limitations.** Gender labels in Bias-in-Bios are binary and inferred from pronoun usage [De-Arteaga et al., 2019], which does not reflect the full spectrum of gender identity. Attribution is restricted to encoder classifiers; IG requires a fixed class-logit target, which does not directly apply to autoregressive Pythia without non-trivial adaptation. Names were not masked; since names are gender-correlated, they may act as direct signals rather than proxies. Attribution findings thus reflect proxy patterns among non-name tokens. Attribution scores are correlational, not causal, indicating statistical reliance rather than causal determination of model behaviour. Our erasure-based faithfulness test and counterfactual pronoun swap provide limited interventional evidence; because token removal alters the input distribution and pronoun swaps leave non-pronominal gender signals intact, results are best interpreted as evidence of functional dependence rather than definitive causal tests. Per-profession sample sizes are small for rare occupations (comedian  $n=27$ , dietitian  $n=30$ ), making per-profession TPR estimates unreliable for those classes. Macro-averaging gives an aggregate EO standard error of  $\approx 0.018$ ; the larger zero-shot gaps ( $\sim 0.09$ – $0.21$ ) are well above this noise floor, while fine-tuned differences remain close to it.

**Future work.** Extending attribution analysis to generative models would explain Pythia’s fairness disparities more directly. Developing reliable uncertainty estimates for the EO gap remains open: standard bootstrap CIs degrade under the  $\max()$  operator and severe class imbalance, so analytical bounds or permutation-based tests may be more appropriate here. Extending counterfactual analysis to other lexical groups, or applying causal mediation, would provide stronger causal evidence.



## Acknowledgements

We thank our teaching assistant Karen Hambarzumyan for his guidance and support throughout the semester.

## A Per-Occupation Proxy Word Tables

Top-5 attributed tokens (weighted score =  $\text{count\_topk} \times \text{mean attribution}$ ) per profession for DistilBERT and RoBERTa under unmasked (U) and masked (M) conditions. Aggregated over 3,000 test examples; stopwords and tokens shorter than 3 characters excluded. *Note: occasional RoBERTa entries may reflect corpus artefacts (e.g. proper nouns or URL fragments from the training data) rather than semantically meaningful profession features.*

### Accountant

| DistilBERT (U/M) |         | RoBERTa (U/M) |         |
|------------------|---------|---------------|---------|
| Token            | W.Score | Token         | W.Score |
| <i>Unmasked</i>  |         |               |         |
| accounting       | 20.8    | accounting    | 21.2    |
| financial        | 9.5     | financial     | 6.7     |
| accountants      | 4.3     | accountants   | 5.3     |
| accounts         | 3.8     | business      | 5.3     |
| finance          | 3.1     | tax           | 5.1     |
| <i>Masked</i>    |         |               |         |
| accounting       | 25.2    | accounting    | 30.7    |
| financial        | 7.3     | financial     | 10.5    |
| accountants      | 4.6     | finance       | 10.2    |
| finance          | 2.8     | business      | 7.4     |
| accounts         | 2.7     | cfo           | 6.2     |

### Architect

| DistilBERT (U/M) |         | RoBERTa (U/M) |         |
|------------------|---------|---------------|---------|
| Token            | W.Score | Token         | W.Score |
| <i>Unmasked</i>  |         |               |         |
| architecture     | 29.7    | architecture  | 38.9    |
| architectural    | 15.5    | architectural | 24.6    |
| design           | 13.1    | design        | 17.7    |
| architect        | 11.1    | architect     | 10.6    |
| architects       | 7.9     | architects    | 10.4    |
| <i>Masked</i>    |         |               |         |
| architecture     | 38.8    | architecture  | 57.1    |
| architectural    | 19.8    | architectural | 29.9    |
| architect        | 12.4    | design        | 24.1    |
| design           | 10.3    | architect     | 12.5    |
| architects       | 6.2     | architects    | 12.2    |

### Attorney

| DistilBERT (U/M) |         | RoBERTa (U/M) |         |
|------------------|---------|---------------|---------|
| Token            | W.Score | Token         | W.Score |
| <i>Unmasked</i>  |         |               |         |
| law              | 56.0    | law           | 143.6   |
| attorney         | 37.4    | litigation    | 54.0    |
| legal            | 27.6    | legal         | 49.5    |
| litigation       | 25.5    | practice      | 47.1    |
| clients          | 19.8    | attorney      | 46.2    |
| <i>Masked</i>    |         |               |         |
| law              | 82.0    | law           | 182.2   |
| attorney         | 33.2    | litigation    | 70.7    |
| litigation       | 30.2    | attorney      | 56.7    |
| legal            | 25.5    | practice      | 52.3    |
| clients          | 18.4    | legal         | 31.7    |

### Comedian

| DistilBERT (U/M) |         | RoBERTa (U/M) |         |
|------------------|---------|---------------|---------|
| Token            | W.Score | Token         | W.Score |
| <i>Unmasked</i>  |         |               |         |
| comedy           | 16.4    | comedy        | 12.6    |
| stand            | 4.3     | comedians     | 4.6     |
| comedians        | 3.0     | stand-up      | 4.4     |
| performed        | 2.0     | joke          | 3.6     |
| april            | 1.3     | comedic       | 3.2     |
| <i>Masked</i>    |         |               |         |
| comedy           | 16.5    | comedy        | 23.7    |
| stand            | 4.5     | bajwa         | 8.4     |
| comedians        | 4.1     | comedians     | 5.0     |
| performed        | 2.3     | stand-up      | 4.9     |
| jokes            | 1.9     | show          | 3.4     |

### Composer

| DistilBERT (U/M) |         | RoBERTa (U/M) |         |
|------------------|---------|---------------|---------|
| Token            | W.Score | Token         | W.Score |
| <i>Unmasked</i>  |         |               |         |
| music            | 24.3    | short         | 25.5    |
| composition      | 6.9     | music         | 22.4    |
| composer         | 6.4     | composition   | 9.6     |
| musical          | 4.6     | compositions  | 7.9     |
| composing        | 4.3     | musical       | 7.1     |
| <i>Masked</i>    |         |               |         |
| music            | 37.4    | music         | 53.2    |
| composition      | 9.1     | composition   | 12.2    |
| composer         | 5.8     | compositions  | 10.9    |
| compositions     | 5.8     | musical       | 9.5     |
| musical          | 5.7     | orchestra     | 8.6     |

### Dentist

| DistilBERT (U/M) |         | RoBERTa (U/M) |         |
|------------------|---------|---------------|---------|
| Token            | W.Score | Token         | W.Score |
| <i>Unmasked</i>  |         |               |         |
| dental           | 107.2   | dental        | 185.2   |
| dentistry        | 48.8    | dentistry     | 39.8    |
| dentist          | 28.9    | clinic        | 25.2    |
| bds              | 23.4    | dentist       | 24.1    |
| clinic           | 9.6     | practices     | 19.6    |
| <i>Masked</i>    |         |               |         |
| dental           | 107.4   | dental        | 200.2   |
| dentistry        | 41.1    | dentistry     | 45.8    |
| dentist          | 32.2    | clinic        | 29.6    |
| patients         | 9.9     | patients      | 27.9    |
| dentists         | 8.8     | dentist       | 27.4    |

### Dietitian

| DistilBERT (U/M)   |         | RoBERTa (U/M)       |         |
|--------------------|---------|---------------------|---------|
| Token              | W.Score | Token               | W.Score |
| <i>Unmasked</i>    |         |                     |         |
| <i>nutrition</i>   | 37.7    | <i>nutrition</i>    | 30.7    |
| <i>dietitian</i>   | 35.3    | <i>dietitian</i>    | 27.8    |
| <i>registered</i>  | 6.4     | <i>professional</i> | 10.4    |
| <i>nutritional</i> | 3.8     | <i>nutritional</i>  | 5.1     |
| <i>diet</i>        | 3.2     | <i>registered</i>   | 4.7     |
| <i>Masked</i>      |         |                     |         |
| <i>dietitian</i>   | 37.5    | <i>nutrition</i>    | 45.7    |
| <i>nutrition</i>   | 35.7    | <i>professional</i> | 21.6    |
| <i>registered</i>  | 7.4     | <i>dietitian</i>    | 14.5    |
| <i>diets</i>       | 4.8     | <i>nutritional</i>  | 5.0     |
| <i>diet</i>        | 4.6     | <i>appointment</i>  | 3.8     |

## Filmmaker

| DistilBERT (U/M)   |         | RoBERTa (U/M)     |         |
|--------------------|---------|-------------------|---------|
| Token              | W.Score | Token             | W.Score |
| <i>Unmasked</i>    |         |                   |         |
| <i>film</i>        | 21.0    | <i>films</i>      | 19.7    |
| <i>films</i>       | 12.5    | <i>film</i>       | 17.1    |
| <i>documentary</i> | 8.5     | <i>filmmaker</i>  | 7.8     |
| <i>filmmaking</i>  | 7.8     | <i>feature</i>    | 7.6     |
| <i>directed</i>    | 4.6     | <i>directed</i>   | 6.0     |
| <i>Masked</i>      |         |                   |         |
| <i>film</i>        | 32.2    | <i>film</i>       | 39.5    |
| <i>films</i>       | 16.4    | <i>films</i>      | 28.8    |
| <i>documentary</i> | 7.5     | <i>filmmaking</i> | 12.0    |
| <i>filmmaking</i>  | 5.8     | <i>filmmaker</i>  | 9.0     |
| <i>directed</i>    | 5.5     | <i>feature</i>    | 8.3     |

## Journalist

| DistilBERT (U/M)   |         | RoBERTa (U/M)      |         |
|--------------------|---------|--------------------|---------|
| Token              | W.Score | Token              | W.Score |
| <i>Unmasked</i>    |         |                    |         |
| <i>journalism</i>  | 34.6    | <i>journalism</i>  | 28.7    |
| <i>journalist</i>  | 18.1    | <i>news</i>        | 16.2    |
| <i>news</i>        | 13.7    | <i>journalist</i>  | 15.4    |
| <i>editor</i>      | 11.5    | <i>reporting</i>   | 8.8     |
| <i>journalists</i> | 8.6     | <i>times</i>       | 8.4     |
| <i>Masked</i>      |         |                    |         |
| <i>journalism</i>  | 27.8    | <i>journalism</i>  | 44.6    |
| <i>journalist</i>  | 16.6    | <i>news</i>        | 24.6    |
| <i>editor</i>      | 12.8    | <i>journalist</i>  | 22.2    |
| <i>news</i>        | 10.6    | <i>journalists</i> | 15.6    |
| <i>reporting</i>   | 8.6     | <i>writing</i>     | 10.4    |

## Model

| DistilBERT (U/M) |         | RoBERTa (U/M)    |         |
|------------------|---------|------------------|---------|
| Token            | W.Score | Token            | W.Score |
| <i>Unmasked</i>  |         |                  |         |
| <i>model</i>     | 17.1    | <i>model</i>     | 24.2    |
| <i>freeones</i>  | 15.0    | <i>modeling</i>  | 11.5    |
| <i>modeling</i>  | 14.2    | <i>ranked</i>    | 10.4    |
| <i>born</i>      | 7.2     | <i>born</i>      | 8.4     |
| <i>gallery</i>   | 5.3     | <i>modelling</i> | 6.0     |
| <i>Masked</i>    |         |                  |         |
| <i>model</i>     | 12.1    | <i>model</i>     | 31.1    |
| <i>modeling</i>  | 11.5    | <i>born</i>      | 22.7    |
| <i>born</i>      | 7.7     | <i>gallery</i>   | 18.1    |
| <i>freeones</i>  | 7.5     | <i>modeling</i>  | 17.1    |
| <i>ranked</i>    | 6.0     | <i>modelling</i> | 8.2     |

## Nurse

| DistilBERT (U/M)    |         | RoBERTa (U/M)       |         |
|---------------------|---------|---------------------|---------|
| Token               | W.Score | Token               | W.Score |
| <i>Unmasked</i>     |         |                     |         |
| <i>nurse</i>        | 129.9   | <i>nurse</i>        | 141.1   |
| <i>nursing</i>      | 55.5    | <i>nursing</i>      | 70.4    |
| <i>practitioner</i> | 26.5    | <i>practitioner</i> | 25.3    |
| <i>graduated</i>    | 19.1    | <i>graduated</i>    | 24.9    |
| <i>experiences</i>  | 15.2    | <i>medical</i>      | 13.3    |
| <i>Masked</i>       |         |                     |         |
| <i>nurse</i>        | 152.7   | <i>nurse</i>        | 166.4   |
| <i>nursing</i>      | 60.7    | <i>nursing</i>      | 85.1    |
| <i>especially</i>   | 15.5    | <i>practitioner</i> | 20.5    |
| <i>practitioner</i> | 11.2    | <i>call</i>         | 16.0    |
| <i>hospital</i>     | 9.0     | <i>medical</i>      | 14.2    |

## Painter

| DistilBERT (U/M) |         | RoBERTa (U/M)    |         |
|------------------|---------|------------------|---------|
| Token            | W.Score | Token            | W.Score |
| <i>Unmasked</i>  |         |                  |         |
| <i>paintings</i> | 25.6    | <i>painting</i>  | 36.1    |
| <i>painting</i>  | 21.0    | <i>paintings</i> | 24.3    |
| <i>gallery</i>   | 8.2     | <i>art</i>       | 8.1     |
| <i>art</i>       | 8.0     | <i>gallery</i>   | 5.4     |
| <i>paint</i>     | 4.1     | <i>paint</i>     | 4.8     |
| <i>Masked</i>    |         |                  |         |
| <i>paintings</i> | 25.7    | <i>painting</i>  | 51.4    |
| <i>painting</i>  | 24.5    | <i>paintings</i> | 50.0    |
| <i>art</i>       | 8.1     | <i>art</i>       | 13.3    |
| <i>gallery</i>   | 4.7     | <i>gallery</i>   | 9.6     |
| <i>work</i>      | 4.4     | <i>depicted</i>  | 8.7     |

## Photographer

| DistilBERT (U/M)    |         | RoBERTa (U/M)       |         |
|---------------------|---------|---------------------|---------|
| Token               | W.Score | Token               | W.Score |
| <i>Unmasked</i>     |         |                     |         |
| <i>photography</i>  | 149.1   | <i>photography</i>  | 126.0   |
| <i>photographs</i>  | 26.2    | <i>photographs</i>  | 23.9    |
| <i>photographer</i> | 23.6    | <i>photographer</i> | 23.0    |
| <i>photographic</i> | 16.2    | <i>photographic</i> | 15.1    |
| <i>photo</i>        | 14.3    | <i>shoots</i>       | 12.8    |
| <i>Masked</i>       |         |                     |         |
| <i>photography</i>  | 143.0   | <i>photography</i>  | 239.6   |
| <i>photographs</i>  | 24.3    | <i>photographs</i>  | 45.2    |
| <i>photographer</i> | 22.0    | <i>photographer</i> | 39.2    |
| <i>work</i>         | 15.9    | <i>photographic</i> | 29.4    |
| <i>photographic</i> | 15.8    | <i>images</i>       | 24.9    |

## Physician

| DistilBERT (U/M) |         | RoBERTa (U/M)      |         |
|------------------|---------|--------------------|---------|
| Token            | W.Score | Token              | W.Score |
| <i>Unmasked</i>  |         |                    |         |
| <i>medicine</i>  | 268.2   | <i>practices</i>   | 179.3   |
| <i>practices</i> | 76.9    | <i>medicine</i>    | 166.6   |
| <i>medical</i>   | 66.4    | <i>medical</i>     | 77.4    |
| <i>physician</i> | 36.1    | <i>affiliated</i>  | 68.8    |
| <i>graduate</i>  | 15.9    | <i>physician</i>   | 26.0    |
| <i>Masked</i>    |         |                    |         |
| <i>medicine</i>  | 287.6   | <i>medicine</i>    | 292.2   |
| <i>medical</i>   | 89.3    | <i>practices</i>   | 224.6   |
| <i>practices</i> | 72.8    | <i>medical</i>     | 97.4    |
| <i>hospital</i>  | 33.0    | <i>patients</i>    | 52.7    |
| <i>physician</i> | 31.0    | <i>specializes</i> | 38.3    |

## Poet

| DistilBERT (U/M) |         | RoBERTa (U/M)    |         |
|------------------|---------|------------------|---------|
| Token            | W.Score | Token            | W.Score |
| <i>Unmasked</i>  |         |                  |         |
| <i>poetry</i>    | 34.9    | <i>poetry</i>    | 52.0    |
| <i>poems</i>     | 8.1     | <i>poems</i>     | 15.7    |
| <i>writing</i>   | 4.7     | <i>writing</i>   | 8.7     |
| <i>english</i>   | 3.1     | <i>poem</i>      | 6.3     |
| <i>poem</i>      | 3.0     | <i>published</i> | 3.6     |
| <i>Masked</i>    |         |                  |         |
| <i>poetry</i>    | 44.9    | <i>poetry</i>    | 67.8    |
| <i>poems</i>     | 9.9     | <i>poems</i>     | 14.0    |
| <i>writing</i>   | 5.1     | <i>writing</i>   | 7.6     |
| <i>poem</i>      | 4.5     | <i>poem</i>      | 6.6     |
| <i>english</i>   | 3.5     | <i>review</i>    | 6.6     |

## Professor

| DistilBERT (U/M)  |         | RoBERTa (U/M)     |         |
|-------------------|---------|-------------------|---------|
| Token             | W.Score | Token             | W.Score |
| <i>Unmasked</i>   |         |                   |         |
| <i>research</i>   | 329.5   | <i>research</i>   | 333.3   |
| <i>phd</i>        | 93.8    | <i>ph.d</i>       | 170.5   |
| <i>interests</i>  | 60.9    | <i>phd</i>        | 119.2   |
| <i>university</i> | 58.6    | <i>university</i> | 105.0   |
| <i>focuses</i>    | 49.3    | <i>interests</i>  | 48.4    |
| <i>Masked</i>     |         |                   |         |
| <i>research</i>   | 377.2   | <i>research</i>   | 514.2   |
| <i>university</i> | 193.5   | <i>university</i> | 302.8   |
| <i>phd</i>        | 79.8    | <i>interests</i>  | 95.4    |
| <i>professor</i>  | 49.0    | <i>phd</i>        | 94.1    |
| <i>interests</i>  | 46.9    | <i>ph.d</i>       | 91.7    |

## Psychologist

| DistilBERT (U/M)     |         | RoBERTa (U/M)        |         |
|----------------------|---------|----------------------|---------|
| Token                | W.Score | Token                | W.Score |
| <i>Unmasked</i>      |         |                      |         |
| <i>psychologist</i>  | 58.6    | <i>psychologist</i>  | 55.0    |
| <i>psychology</i>    | 43.0    | <i>psychology</i>    | 31.5    |
| <i>psychological</i> | 9.2     | <i>clinical</i>      | 18.7    |
| <i>psychotherapy</i> | 9.2     | <i>appointment</i>   | 13.2    |
| <i>experiences</i>   | 8.3     | <i>experiences</i>   | 10.5    |
| <i>Masked</i>        |         |                      |         |
| <i>psychologist</i>  | 56.4    | <i>psychologist</i>  | 61.6    |
| <i>psychology</i>    | 33.0    | <i>psychology</i>    | 40.4    |
| <i>clinical</i>      | 17.5    | <i>clinical</i>      | 22.0    |
| <i>psychological</i> | 8.4     | <i>psychological</i> | 16.8    |
| <i>psychotherapy</i> | 7.1     | <i>psychotherapy</i> | 13.4    |

## Software Engineer

| DistilBERT (U/M)   |         | RoBERTa (U/M)      |         |
|--------------------|---------|--------------------|---------|
| Token              | W.Score | Token              | W.Score |
| <i>Unmasked</i>    |         |                    |         |
| <i>software</i>    | 4.8     | <i>development</i> | 12.3    |
| <i>development</i> | 4.6     | <i>engineering</i> | 9.7     |
| <i>engineering</i> | 4.2     | <i>computer</i>    | 6.4     |
| <i>ibm</i>         | 2.9     | <i>software</i>    | 6.3     |
| <i>code</i>        | 2.6     | <i>building</i>    | 3.4     |
| <i>Masked</i>      |         |                    |         |
| <i>computer</i>    | 7.4     | <i>software</i>    | 14.2    |
| <i>engineering</i> | 7.4     | <i>engineering</i> | 8.0     |
| <i>software</i>    | 5.8     | <i>development</i> | 4.7     |
| <i>development</i> | 4.5     | <i>javascript</i>  | 4.5     |
| <i>intel</i>       | 3.7     | <i>devops</i>      | 4.4     |

## Surgeon

| DistilBERT (U/M)   |         | RoBERTa (U/M)       |         |
|--------------------|---------|---------------------|---------|
| Token              | W.Score | Token               | W.Score |
| <i>Unmasked</i>    |         |                     |         |
| <i>surgery</i>     | 44.2    | <i>surgery</i>      | 51.1    |
| <i>medical</i>     | 13.2    | <i>medical</i>      | 20.0    |
| <i>medicine</i>    | 12.1    | <i>practice</i>     | 16.5    |
| <i>practices</i>   | 8.9     | <i>medicine</i>     | 12.2    |
| <i>surgeon</i>     | 7.7     | <i>hospital</i>     | 10.8    |
| <i>Masked</i>      |         |                     |         |
| <i>surgery</i>     | 24.4    | <i>surgery</i>      | 95.6    |
| <i>medical</i>     | 15.9    | <i>medical</i>      | 22.9    |
| <i>practice</i>    | 12.7    | <i>surgeon</i>      | 16.8    |
| <i>medicine</i>    | 11.7    | <i>medicine</i>     | 13.9    |
| <i>orthopaedic</i> | 8.0     | <i>neurosurgery</i> | 10.6    |

## Teacher

| DistilBERT (U/M) |         | RoBERTa (U/M)   |         |
|------------------|---------|-----------------|---------|
| Token            | W.Score | Token           | W.Score |
| <i>Unmasked</i>  |         |                 |         |
| <i>teacher</i>   | 19.9    | <i>students</i> | 25.3    |
| <i>students</i>  | 18.7    | <i>teaching</i> | 20.7    |
| <i>teaching</i>  | 16.3    | <i>teacher</i>  | 20.4    |
| <i>school</i>    | 12.8    | <i>school</i>   | 18.6    |
| <i>teachers</i>  | 12.4    | <i>taught</i>   | 13.7    |
| <i>Masked</i>    |         |                 |         |
| <i>students</i>  | 27.1    | <i>students</i> | 38.4    |
| <i>teacher</i>   | 18.2    | <i>teaching</i> | 34.2    |
| <i>teaching</i>  | 14.5    | <i>school</i>   | 26.3    |
| <i>teachers</i>  | 12.3    | <i>teaches</i>  | 26.1    |
| <i>school</i>    | 10.8    | <i>teacher</i>  | 23.6    |

## B Prompt Templates

### B.1 Zero-Shot

```
"""
Given the short biography below, predict the occupation of the person.
Answer with a single specific occupation word (e.g. professor,
physician, attorney, photographer, journalist, nurse, psychologist,
teacher, dentist, surgeon, architect, painter, model, poet, filmmaker,
software_engineer, accountant, composer, dietitian, comedian etc.).

```

Biography: {text}

Occupation:

```
"""
```

### B.2 Few-Shot

```
"""
Given the short biography below, predict the occupation of the person.
Answer with a single specific occupation word (e.g. professor, physician,
attorney, photographer, journalist, nurse, psychologist, teacher, dentist,
surgeon, architect, painter, model, poet, filmmaker, software_engineer,
accountant, composer, dietitian, comedian etc.).

```

Biography: She is a renowned researcher and teaches several courses at the university.

Occupation: professor

Biography: The patient was admitted to the hospital, where he was treated by the attending doctor.

Occupation: physician

Biography: He covers local news and politics for the daily newspaper.

Occupation: journalist

Biography: She has been representing clients in corporate litigation for over a decade.

Occupation: attorney

Biography: {text}

Occupation:

```
"""
```

## C Detailed Fairness Metrics by Occupation

For the 3000 test samples, the EO and DP gap for every occupation and every model, for both masked and unmasked cases.

### Accountant



| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.094 | 0.002 | 0.012 | Pythia-410m FT | M  | 0.042 | 0.001 | 0.003 |
| DistilBERT FT  | M  | 0.031 | 0.000 | 0.009 | Pythia-410m FS | U  | 0.281 | 0.007 | 0.015 |
| RoBERTa FT     | U  | 0.062 | 0.000 | 0.009 | Pythia-410m FS | M  | 0.135 | 0.003 | 0.010 |
| RoBERTa FT     | M  | 0.010 | 0.001 | 0.008 | Pythia-410m 0s | U  | 0.010 | 0.000 | 0.004 |
| Pythia-1.4B FT | U  | 0.229 | 0.002 | 0.013 | Pythia-410m 0s | M  | 0.167 | 0.002 | 0.009 |
| Pythia-1.4B FT | M  | 0.229 | 0.002 | 0.013 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.344 | 0.007 | 0.016 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.198 | 0.000 | 0.008 | Pythia-160m FS | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B 0s | U  | 0.260 | 0.001 | 0.001 | Pythia-160m FS | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B 0s | M  | 0.344 | 0.000 | 0.001 | Pythia-160m 0s | U  | 0.000 | 0.000 | 0.000 |
| Pythia-410m FT | U  | 0.062 | 0.003 | 0.001 | Pythia-160m 0s | M  | 0.000 | 0.000 | 0.000 |

## Architect

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.242 | 0.006 | 0.017 | Pythia-410m FT | M  | 0.028 | 0.002 | 0.005 |
| DistilBERT FT  | M  | 0.134 | 0.008 | 0.019 | Pythia-410m FS | U  | 0.012 | 0.015 | 0.024 |
| RoBERTa FT     | U  | 0.173 | 0.008 | 0.020 | Pythia-410m FS | M  | 0.038 | 0.010 | 0.018 |
| RoBERTa FT     | M  | 0.156 | 0.007 | 0.020 | Pythia-410m 0s | U  | 0.099 | 0.064 | 0.081 |
| Pythia-1.4B FT | U  | 0.209 | 0.005 | 0.017 | Pythia-410m 0s | M  | 0.032 | 0.037 | 0.052 |
| Pythia-1.4B FT | M  | 0.175 | 0.003 | 0.016 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.026 | 0.014 | 0.024 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.026 | 0.009 | 0.020 | Pythia-160m FS | U  | 0.131 | 0.003 | 0.009 |
| Pythia-1.4B 0s | U  | 0.135 | 0.002 | 0.005 | Pythia-160m FS | M  | 0.009 | 0.001 | 0.004 |
| Pythia-1.4B 0s | M  | 0.168 | 0.002 | 0.005 | Pythia-160m 0s | U  | 0.003 | 0.001 | 0.002 |
| Pythia-410m FT | U  | 0.182 | 0.015 | 0.015 | Pythia-160m 0s | M  | 0.072 | 0.002 | 0.002 |

## Attorney

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.046 | 0.001 | 0.019 | Pythia-410m FT | M  | 0.042 | 0.004 | 0.010 |
| DistilBERT FT  | M  | 0.002 | 0.000 | 0.016 | Pythia-410m FS | U  | 0.070 | 0.014 | 0.029 |
| RoBERTa FT     | U  | 0.026 | 0.000 | 0.018 | Pythia-410m FS | M  | 0.034 | 0.015 | 0.027 |
| RoBERTa FT     | M  | 0.030 | 0.003 | 0.016 | Pythia-410m 0s | U  | 0.198 | 0.009 | 0.038 |
| Pythia-1.4B FT | U  | 0.014 | 0.004 | 0.014 | Pythia-410m 0s | M  | 0.100 | 0.006 | 0.028 |
| Pythia-1.4B FT | M  | 0.020 | 0.001 | 0.017 | Pythia-160m FT | U  | 0.026 | 0.022 | 0.024 |
| Pythia-1.4B FS | U  | 0.029 | 0.030 | 0.012 | Pythia-160m FT | M  | 0.021 | 0.015 | 0.017 |
| Pythia-1.4B FS | M  | 0.066 | 0.012 | 0.006 | Pythia-160m FS | U  | 0.122 | 0.002 | 0.013 |
| Pythia-1.4B 0s | U  | 0.055 | 0.005 | 0.001 | Pythia-160m FS | M  | 0.079 | 0.003 | 0.009 |
| Pythia-1.4B 0s | M  | 0.040 | 0.001 | 0.011 | Pythia-160m 0s | U  | 0.017 | 0.000 | 0.001 |
| Pythia-410m FT | U  | 0.009 | 0.005 | 0.015 | Pythia-160m 0s | M  | 0.009 | 0.001 | 0.004 |

## Comedian

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.059 | 0.004 | 0.010 | Pythia-410m FT | M  | 0.000 | 0.002 | 0.002 |
| DistilBERT FT  | M  | 0.118 | 0.002 | 0.008 | Pythia-410m FS | U  | 0.588 | 0.001 | 0.007 |
| RoBERTa FT     | U  | 0.141 | 0.003 | 0.010 | Pythia-410m FS | M  | 0.212 | 0.000 | 0.004 |
| RoBERTa FT     | M  | 0.141 | 0.002 | 0.009 | Pythia-410m 0s | U  | 0.271 | 0.001 | 0.005 |
| Pythia-1.4B FT | U  | 0.141 | 0.001 | 0.008 | Pythia-410m 0s | M  | 0.247 | 0.001 | 0.002 |
| Pythia-1.4B FT | M  | 0.141 | 0.000 | 0.007 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.047 | 0.000 | 0.005 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.106 | 0.000 | 0.006 | Pythia-160m FS | U  | 0.118 | 0.001 | 0.000 |
| Pythia-1.4B 0s | U  | 0.047 | 0.000 | 0.002 | Pythia-160m FS | M  | 0.082 | 0.001 | 0.001 |
| Pythia-1.4B 0s | M  | 0.294 | 0.001 | 0.004 | Pythia-160m 0s | U  | 0.235 | 0.000 | 0.002 |
| Pythia-410m FT | U  | 0.000 | 0.002 | 0.002 | Pythia-160m 0s | M  | 0.176 | 0.001 | 0.002 |

## Composer

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.025 | 0.001 | 0.015 | Pythia-410m FT | M  | 0.025 | 0.001 | 0.002 |
| DistilBERT FT  | M  | 0.025 | 0.001 | 0.015 | Pythia-410m FS | U  | 0.152 | 0.001 | 0.009 |
| RoBERTa FT     | U  | 0.003 | 0.000 | 0.013 | Pythia-410m FS | M  | 0.152 | 0.002 | 0.010 |
| RoBERTa FT     | M  | 0.054 | 0.001 | 0.014 | Pythia-410m 0s | U  | 0.003 | 0.003 | 0.016 |
| Pythia-1.4B FT | U  | 0.025 | 0.001 | 0.015 | Pythia-410m 0s | M  | 0.060 | 0.003 | 0.015 |
| Pythia-1.4B FT | M  | 0.054 | 0.000 | 0.015 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.117 | 0.004 | 0.014 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.006 | 0.001 | 0.012 | Pythia-160m FS | U  | 0.060 | 0.003 | 0.000 |
| Pythia-1.4B 0s | U  | 0.067 | 0.003 | 0.011 | Pythia-160m FS | M  | 0.063 | 0.001 | 0.006 |
| Pythia-1.4B 0s | M  | 0.038 | 0.001 | 0.010 | Pythia-160m 0s | U  | 0.006 | 0.001 | 0.003 |
| Pythia-410m FT | U  | 0.165 | 0.004 | 0.004 | Pythia-160m 0s | M  | 0.051 | 0.000 | 0.002 |

Dentist

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.057 | 0.002 | 0.030 | Pythia-410m FT | M  | 0.053 | 0.001 | 0.020 |
| DistilBERT FT  | M  | 0.080 | 0.002 | 0.029 | Pythia-410m FS | U  | 0.075 | 0.001 | 0.024 |
| RoBERTa FT     | U  | 0.068 | 0.002 | 0.029 | Pythia-410m FS | M  | 0.119 | 0.003 | 0.023 |
| RoBERTa FT     | M  | 0.057 | 0.002 | 0.030 | Pythia-410m 0s | U  | 0.006 | 0.002 | 0.026 |
| Pythia-1.4B FT | U  | 0.068 | 0.002 | 0.029 | Pythia-410m 0s | M  | 0.031 | 0.005 | 0.021 |
| Pythia-1.4B FT | M  | 0.068 | 0.002 | 0.029 | Pythia-160m FT | U  | 0.194 | 0.002 | 0.006 |
| Pythia-1.4B FS | U  | 0.039 | 0.006 | 0.030 | Pythia-160m FT | M  | 0.057 | 0.001 | 0.002 |
| Pythia-1.4B FS | M  | 0.004 | 0.002 | 0.025 | Pythia-160m FS | U  | 0.053 | 0.000 | 0.021 |
| Pythia-1.4B 0s | U  | 0.019 | 0.001 | 0.028 | Pythia-160m FS | M  | 0.044 | 0.004 | 0.013 |
| Pythia-1.4B 0s | M  | 0.019 | 0.005 | 0.023 | Pythia-160m 0s | U  | 0.001 | 0.000 | 0.016 |
| Pythia-410m FT | U  | 0.082 | 0.002 | 0.019 | Pythia-160m 0s | M  | 0.010 | 0.001 | 0.015 |

Dietitian

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.103 | 0.000 | 0.018 | Pythia-410m FT | M  | 0.034 | 0.003 | 0.004 |
| DistilBERT FT  | M  | 0.034 | 0.000 | 0.019 | Pythia-410m FS | U  | 0.017 | 0.001 | 0.011 |
| RoBERTa FT     | U  | 0.069 | 0.001 | 0.019 | Pythia-410m FS | M  | 0.259 | 0.005 | 0.009 |
| RoBERTa FT     | M  | 0.103 | 0.000 | 0.018 | Pythia-410m 0s | U  | 0.552 | 0.001 | 0.012 |
| Pythia-1.4B FT | U  | 0.069 | 0.001 | 0.019 | Pythia-410m 0s | M  | 0.086 | 0.002 | 0.010 |
| Pythia-1.4B FT | M  | 0.069 | 0.000 | 0.018 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.086 | 0.009 | 0.020 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.121 | 0.016 | 0.027 | Pythia-160m FS | U  | 0.138 | 0.005 | 0.007 |
| Pythia-1.4B 0s | U  | 0.862 | 0.000 | 0.018 | Pythia-160m FS | M  | 0.103 | 0.004 | 0.006 |
| Pythia-1.4B 0s | M  | 0.483 | 0.001 | 0.011 | Pythia-160m 0s | U  | 0.000 | 0.001 | 0.001 |
| Pythia-410m FT | U  | 0.034 | 0.006 | 0.007 | Pythia-160m 0s | M  | 0.000 | 0.000 | 0.000 |

Filmmaker

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.079 | 0.003 | 0.007 | Pythia-410m FT | M  | 0.029 | 0.002 | 0.002 |
| DistilBERT FT  | M  | 0.021 | 0.003 | 0.009 | Pythia-410m FS | U  | 0.014 | 0.000 | 0.002 |
| RoBERTa FT     | U  | 0.064 | 0.003 | 0.008 | Pythia-410m FS | M  | 0.086 | 0.002 | 0.003 |
| RoBERTa FT     | M  | 0.043 | 0.003 | 0.009 | Pythia-410m 0s | U  | 0.207 | 0.001 | 0.005 |
| Pythia-1.4B FT | U  | 0.036 | 0.001 | 0.008 | Pythia-410m 0s | M  | 0.179 | 0.001 | 0.005 |
| Pythia-1.4B FT | M  | 0.014 | 0.002 | 0.009 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.036 | 0.001 | 0.005 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.057 | 0.000 | 0.004 | Pythia-160m FS | U  | 0.121 | 0.001 | 0.004 |
| Pythia-1.4B 0s | U  | 0.071 | 0.000 | 0.001 | Pythia-160m FS | M  | 0.043 | 0.000 | 0.001 |
| Pythia-1.4B 0s | M  | 0.071 | 0.000 | 0.001 | Pythia-160m 0s | U  | 0.000 | 0.000 | 0.000 |
| Pythia-410m FT | U  | 0.029 | 0.003 | 0.003 | Pythia-160m 0s | M  | 0.021 | 0.000 | 0.000 |

Journalist

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.076 | 0.003 | 0.012 | Pythia-410m FT | M  | 0.043 | 0.006 | 0.002 |
| DistilBERT FT  | M  | 0.063 | 0.002 | 0.013 | Pythia-410m FS | U  | 0.052 | 0.016 | 0.016 |
| RoBERTa FT     | U  | 0.111 | 0.001 | 0.019 | Pythia-410m FS | M  | 0.073 | 0.006 | 0.006 |
| RoBERTa FT     | M  | 0.048 | 0.004 | 0.010 | Pythia-410m 0s | U  | 0.037 | 0.021 | 0.028 |
| Pythia-1.4B FT | U  | 0.055 | 0.002 | 0.013 | Pythia-410m 0s | M  | 0.032 | 0.002 | 0.006 |
| Pythia-1.4B FT | M  | 0.033 | 0.007 | 0.006 | Pythia-160m FT | U  | 0.030 | 0.004 | 0.002 |
| Pythia-1.4B FS | U  | 0.119 | 0.002 | 0.002 | Pythia-160m FT | M  | 0.024 | 0.000 | 0.002 |
| Pythia-1.4B FS | M  | 0.042 | 0.002 | 0.007 | Pythia-160m FS | U  | 0.111 | 0.013 | 0.013 |
| Pythia-1.4B 0s | U  | 0.350 | 0.004 | 0.028 | Pythia-160m FS | M  | 0.039 | 0.013 | 0.005 |
| Pythia-1.4B 0s | M  | 0.126 | 0.002 | 0.013 | Pythia-160m 0s | U  | 0.074 | 0.004 | 0.009 |
| Pythia-410m FT | U  | 0.009 | 0.002 | 0.003 | Pythia-160m 0s | M  | 0.012 | 0.007 | 0.006 |

Model

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.173 | 0.002 | 0.028 | Pythia-410m FT | M  | 0.380 | 0.003 | 0.010 |
| DistilBERT FT  | M  | 0.304 | 0.000 | 0.028 | Pythia-410m FS | U  | 0.104 | 0.002 | 0.019 |
| RoBERTa FT     | U  | 0.153 | 0.002 | 0.023 | Pythia-410m FS | M  | 0.107 | 0.019 | 0.006 |
| RoBERTa FT     | M  | 0.062 | 0.001 | 0.027 | Pythia-410m 0s | U  | 0.227 | 0.001 | 0.011 |
| Pythia-1.4B FT | U  | 0.173 | 0.001 | 0.027 | Pythia-410m 0s | M  | 0.136 | 0.004 | 0.008 |
| Pythia-1.4B FT | M  | 0.082 | 0.001 | 0.027 | Pythia-160m FT | U  | 0.000 | 0.002 | 0.001 |
| Pythia-1.4B FS | U  | 0.324 | 0.018 | 0.044 | Pythia-160m FT | M  | 0.000 | 0.002 | 0.001 |
| Pythia-1.4B FS | M  | 0.269 | 0.011 | 0.003 | Pythia-160m FS | U  | 0.109 | 0.026 | 0.028 |
| Pythia-1.4B 0s | U  | 0.067 | 0.004 | 0.009 | Pythia-160m FS | M  | 0.224 | 0.005 | 0.004 |
| Pythia-1.4B 0s | M  | 0.073 | 0.000 | 0.008 | Pythia-160m 0s | U  | 0.051 | 0.002 | 0.001 |
| Pythia-410m FT | U  | 0.209 | 0.003 | 0.008 | Pythia-160m 0s | M  | 0.091 | 0.008 | 0.008 |

## Nurse

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.060 | 0.012 | 0.086 | Pythia-410m FT | M  | 0.232 | 0.004 | 0.037 |
| DistilBERT FT  | M  | 0.074 | 0.006 | 0.078 | Pythia-410m FS | U  | 0.007 | 0.050 | 0.077 |
| RoBERTa FT     | U  | 0.067 | 0.009 | 0.084 | Pythia-410m FS | M  | 0.106 | 0.009 | 0.030 |
| RoBERTa FT     | M  | 0.019 | 0.011 | 0.082 | Pythia-410m 0s | U  | 0.049 | 0.020 | 0.083 |
| Pythia-1.4B FT | U  | 0.100 | 0.012 | 0.086 | Pythia-410m 0s | M  | 0.055 | 0.007 | 0.071 |
| Pythia-1.4B FT | M  | 0.034 | 0.009 | 0.079 | Pythia-160m FT | U  | 0.008 | 0.049 | 0.077 |
| Pythia-1.4B FS | U  | 0.092 | 0.035 | 0.073 | Pythia-160m FT | M  | 0.006 | 0.003 | 0.010 |
| Pythia-1.4B FS | M  | 0.095 | 0.009 | 0.040 | Pythia-160m FS | U  | 0.069 | 0.003 | 0.009 |
| Pythia-1.4B 0s | U  | 0.660 | 0.015 | 0.086 | Pythia-160m FS | M  | 0.107 | 0.008 | 0.017 |
| Pythia-1.4B 0s | M  | 0.300 | 0.005 | 0.061 | Pythia-160m 0s | U  | 0.275 | 0.009 | 0.034 |
| Pythia-410m FT | U  | 0.113 | 0.005 | 0.040 | Pythia-160m 0s | M  | 0.051 | 0.004 | 0.013 |

## Painter

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.178 | 0.001 | 0.005 | Pythia-410m FT | M  | 0.018 | 0.004 | 0.004 |
| DistilBERT FT  | M  | 0.046 | 0.002 | 0.000 | Pythia-410m FS | U  | 0.087 | 0.002 | 0.000 |
| RoBERTa FT     | U  | 0.057 | 0.003 | 0.002 | Pythia-410m FS | M  | 0.044 | 0.003 | 0.004 |
| RoBERTa FT     | M  | 0.046 | 0.001 | 0.000 | Pythia-410m 0s | U  | 0.030 | 0.000 | 0.001 |
| Pythia-1.4B FT | U  | 0.043 | 0.000 | 0.002 | Pythia-410m 0s | M  | 0.040 | 0.001 | 0.001 |
| Pythia-1.4B FT | M  | 0.078 | 0.001 | 0.001 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.070 | 0.002 | 0.000 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.101 | 0.000 | 0.002 | Pythia-160m FS | U  | 0.053 | 0.001 | 0.002 |
| Pythia-1.4B 0s | U  | 0.105 | 0.003 | 0.000 | Pythia-160m FS | M  | 0.085 | 0.001 | 0.001 |
| Pythia-1.4B 0s | M  | 0.052 | 0.002 | 0.002 | Pythia-160m 0s | U  | 0.127 | 0.018 | 0.014 |
| Pythia-410m FT | U  | 0.059 | 0.003 | 0.004 | Pythia-160m 0s | M  | 0.020 | 0.001 | 0.002 |

## Photographer

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.013 | 0.001 | 0.023 | Pythia-410m FT | M  | 0.086 | 0.022 | 0.041 |
| DistilBERT FT  | M  | 0.001 | 0.002 | 0.022 | Pythia-410m FS | U  | 0.007 | 0.025 | 0.038 |
| RoBERTa FT     | U  | 0.054 | 0.002 | 0.029 | Pythia-410m FS | M  | 0.025 | 0.027 | 0.014 |
| RoBERTa FT     | M  | 0.030 | 0.002 | 0.023 | Pythia-410m 0s | U  | 0.031 | 0.002 | 0.023 |
| Pythia-1.4B FT | U  | 0.065 | 0.000 | 0.026 | Pythia-410m 0s | M  | 0.029 | 0.000 | 0.017 |
| Pythia-1.4B FT | M  | 0.028 | 0.000 | 0.025 | Pythia-160m FT | U  | 0.081 | 0.024 | 0.030 |
| Pythia-1.4B FS | U  | 0.003 | 0.009 | 0.026 | Pythia-160m FT | M  | 0.081 | 0.004 | 0.004 |
| Pythia-1.4B FS | M  | 0.066 | 0.023 | 0.010 | Pythia-160m FS | U  | 0.003 | 0.015 | 0.001 |
| Pythia-1.4B 0s | U  | 0.057 | 0.009 | 0.008 | Pythia-160m FS | M  | 0.048 | 0.013 | 0.005 |
| Pythia-1.4B 0s | M  | 0.032 | 0.001 | 0.016 | Pythia-160m 0s | U  | 0.069 | 0.001 | 0.003 |
| Pythia-410m FT | U  | 0.046 | 0.005 | 0.024 | Pythia-160m 0s | M  | 0.013 | 0.001 | 0.007 |

## Physician

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.087 | 0.006 | 0.027 | Pythia-410m FT | M  | 0.129 | 0.015 | 0.018 |
| DistilBERT FT  | M  | 0.113 | 0.002 | 0.032 | Pythia-410m FS | U  | 0.079 | 0.011 | 0.006 |
| RoBERTa FT     | U  | 0.087 | 0.001 | 0.031 | Pythia-410m FS | M  | 0.105 | 0.007 | 0.031 |
| RoBERTa FT     | M  | 0.076 | 0.003 | 0.028 | Pythia-410m 0s | U  | 0.141 | 0.047 | 0.039 |
| Pythia-1.4B FT | U  | 0.071 | 0.003 | 0.027 | Pythia-410m 0s | M  | 0.005 | 0.037 | 0.013 |
| Pythia-1.4B FT | M  | 0.083 | 0.002 | 0.029 | Pythia-160m FT | U  | 0.331 | 0.008 | 0.038 |
| Pythia-1.4B FS | U  | 0.092 | 0.004 | 0.022 | Pythia-160m FT | M  | 0.326 | 0.012 | 0.035 |
| Pythia-1.4B FS | M  | 0.079 | 0.002 | 0.026 | Pythia-160m FS | U  | 0.004 | 0.023 | 0.035 |
| Pythia-1.4B 0s | U  | 0.151 | 0.015 | 0.021 | Pythia-160m FS | M  | 0.031 | 0.026 | 0.039 |
| Pythia-1.4B 0s | M  | 0.073 | 0.027 | 0.000 | Pythia-160m 0s | U  | 0.175 | 0.015 | 0.001 |
| Pythia-410m FT | U  | 0.104 | 0.013 | 0.018 | Pythia-160m 0s | M  | 0.199 | 0.035 | 0.014 |

## Poet

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.220 | 0.000 | 0.004 | Pythia-410m FT | M  | 0.127 | 0.004 | 0.002 |
| DistilBERT FT  | M  | 0.153 | 0.002 | 0.004 | Pythia-410m FS | U  | 0.287 | 0.001 | 0.006 |
| RoBERTa FT     | U  | 0.153 | 0.001 | 0.003 | Pythia-410m FS | M  | 0.173 | 0.000 | 0.003 |
| RoBERTa FT     | M  | 0.120 | 0.002 | 0.004 | Pythia-410m 0s | U  | 0.113 | 0.000 | 0.002 |
| Pythia-1.4B FT | U  | 0.120 | 0.001 | 0.003 | Pythia-410m 0s | M  | 0.173 | 0.001 | 0.004 |
| Pythia-1.4B FT | M  | 0.187 | 0.002 | 0.005 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.300 | 0.000 | 0.005 | Pythia-160m FT | M  | 0.000 | 0.002 | 0.002 |
| Pythia-1.4B FS | M  | 0.173 | 0.001 | 0.002 | Pythia-160m FS | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B 0s | U  | 0.133 | 0.002 | 0.004 | Pythia-160m FS | M  | 0.207 | 0.002 | 0.001 |
| Pythia-1.4B 0s | M  | 0.147 | 0.002 | 0.004 | Pythia-160m 0s | U  | 0.020 | 0.007 | 0.007 |
| Pythia-410m FT | U  | 0.060 | 0.001 | 0.002 | Pythia-160m 0s | M  | 0.067 | 0.001 | 0.002 |

## Professor

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.024 | 0.012 | 0.022 | Pythia-410m FT | M  | 0.020 | 0.026 | 0.008 |
| DistilBERT FT  | M  | 0.022 | 0.024 | 0.031 | Pythia-410m FS | U  | 0.056 | 0.010 | 0.027 |
| RoBERTa FT     | U  | 0.018 | 0.024 | 0.032 | Pythia-410m FS | M  | 0.048 | 0.002 | 0.019 |
| RoBERTa FT     | M  | 0.020 | 0.019 | 0.028 | Pythia-410m 0s | U  | 0.299 | 0.034 | 0.121 |
| Pythia-1.4B FT | U  | 0.003 | 0.012 | 0.030 | Pythia-410m 0s | M  | 0.120 | 0.008 | 0.046 |
| Pythia-1.4B FT | M  | 0.003 | 0.002 | 0.023 | Pythia-160m FT | U  | 0.004 | 0.050 | 0.041 |
| Pythia-1.4B FS | U  | 0.143 | 0.013 | 0.059 | Pythia-160m FT | M  | 0.008 | 0.030 | 0.027 |
| Pythia-1.4B FS | M  | 0.062 | 0.013 | 0.035 | Pythia-160m FS | U  | 0.090 | 0.054 | 0.072 |
| Pythia-1.4B 0s | U  | 0.080 | 0.240 | 0.204 | Pythia-160m FS | M  | 0.079 | 0.014 | 0.040 |
| Pythia-1.4B 0s | M  | 0.020 | 0.025 | 0.035 | Pythia-160m 0s | U  | 0.165 | 0.232 | 0.218 |
| Pythia-410m FT | U  | 0.020 | 0.026 | 0.007 | Pythia-160m 0s | M  | 0.074 | 0.058 | 0.069 |

## Psychologist

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.006 | 0.002 | 0.023 | Pythia-410m FT | M  | 0.120 | 0.006 | 0.017 |
| DistilBERT FT  | M  | 0.067 | 0.003 | 0.021 | Pythia-410m FS | U  | 0.037 | 0.007 | 0.014 |
| RoBERTa FT     | U  | 0.034 | 0.000 | 0.024 | Pythia-410m FS | M  | 0.067 | 0.006 | 0.014 |
| RoBERTa FT     | M  | 0.002 | 0.001 | 0.024 | Pythia-410m 0s | U  | 0.024 | 0.004 | 0.014 |
| Pythia-1.4B FT | U  | 0.060 | 0.002 | 0.025 | Pythia-410m 0s | M  | 0.036 | 0.006 | 0.019 |
| Pythia-1.4B FT | M  | 0.031 | 0.002 | 0.024 | Pythia-160m FT | U  | 0.000 | 0.001 | 0.001 |
| Pythia-1.4B FS | U  | 0.084 | 0.015 | 0.030 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.041 | 0.011 | 0.022 | Pythia-160m FS | U  | 0.067 | 0.000 | 0.005 |
| Pythia-1.4B 0s | U  | 0.300 | 0.006 | 0.029 | Pythia-160m FS | M  | 0.032 | 0.002 | 0.001 |
| Pythia-1.4B 0s | M  | 0.019 | 0.001 | 0.012 | Pythia-160m 0s | U  | 0.008 | 0.000 | 0.001 |
| Pythia-410m FT | U  | 0.175 | 0.011 | 0.025 | Pythia-160m 0s | M  | 0.024 | 0.000 | 0.001 |

## Software Engineer

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.070 | 0.010 | 0.028 | Pythia-410m FT | M  | 0.043 | 0.000 | 0.001 |
| DistilBERT FT  | M  | 0.070 | 0.007 | 0.025 | Pythia-410m FS | U  | 0.106 | 0.000 | 0.002 |
| RoBERTa FT     | U  | 0.112 | 0.009 | 0.026 | Pythia-410m FS | M  | 0.079 | 0.019 | 0.019 |
| RoBERTa FT     | M  | 0.134 | 0.008 | 0.024 | Pythia-410m 0s | U  | 0.134 | 0.006 | 0.013 |
| Pythia-1.4B FT | U  | 0.027 | 0.009 | 0.029 | Pythia-410m 0s | M  | 0.274 | 0.010 | 0.015 |
| Pythia-1.4B FT | M  | 0.036 | 0.007 | 0.029 | Pythia-160m FT | U  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | U  | 0.295 | 0.016 | 0.034 | Pythia-160m FT | M  | 0.000 | 0.000 | 0.000 |
| Pythia-1.4B FS | M  | 0.046 | 0.028 | 0.041 | Pythia-160m FS | U  | 0.021 | 0.002 | 0.002 |
| Pythia-1.4B 0s | U  | 0.125 | 0.014 | 0.027 | Pythia-160m FS | M  | 0.043 | 0.004 | 0.005 |
| Pythia-1.4B 0s | M  | 0.103 | 0.017 | 0.029 | Pythia-160m 0s | U  | 0.000 | 0.001 | 0.001 |
| Pythia-410m FT | U  | 0.106 | 0.001 | 0.004 | Pythia-160m 0s | M  | 0.000 | 0.002 | 0.002 |

## Surgeon

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.070 | 0.010 | 0.041 | Pythia-410m FT | M  | 0.252 | 0.005 | 0.024 |
| DistilBERT FT  | M  | 0.020 | 0.007 | 0.037 | Pythia-410m FS | U  | 0.005 | 0.002 | 0.010 |
| RoBERTa FT     | U  | 0.043 | 0.011 | 0.042 | Pythia-410m FS | M  | 0.132 | 0.000 | 0.007 |
| RoBERTa FT     | M  | 0.018 | 0.007 | 0.037 | Pythia-410m 0s | U  | 0.007 | 0.001 | 0.009 |
| Pythia-1.4B FT | U  | 0.116 | 0.012 | 0.045 | Pythia-410m 0s | M  | 0.010 | 0.004 | 0.016 |
| Pythia-1.4B FT | M  | 0.055 | 0.009 | 0.041 | Pythia-160m FT | U  | 0.176 | 0.007 | 0.020 |
| Pythia-1.4B FS | U  | 0.075 | 0.014 | 0.031 | Pythia-160m FT | M  | 0.011 | 0.001 | 0.002 |
| Pythia-1.4B FS | M  | 0.078 | 0.005 | 0.016 | Pythia-160m FS | U  | 0.016 | 0.001 | 0.002 |
| Pythia-1.4B 0s | U  | 0.104 | 0.001 | 0.001 | Pythia-160m FS | M  | 0.027 | 0.001 | 0.000 |
| Pythia-1.4B 0s | M  | 0.093 | 0.002 | 0.003 | Pythia-160m 0s | U  | 0.000 | 0.000 | 0.000 |
| Pythia-410m FT | U  | 0.198 | 0.011 | 0.030 | Pythia-160m 0s | M  | 0.011 | 0.000 | 0.001 |

## Teacher

| Model          | C. | TPR   | FPR   | DP    | Model          | C. | TPR   | FPR   | DP    |
|----------------|----|-------|-------|-------|----------------|----|-------|-------|-------|
| DistilBERT FT  | U  | 0.073 | 0.000 | 0.021 | Pythia-410m FT | M  | 0.053 | 0.010 | 0.017 |
| DistilBERT FT  | M  | 0.040 | 0.006 | 0.025 | Pythia-410m FS | U  | 0.121 | 0.040 | 0.052 |
| RoBERTa FT     | U  | 0.132 | 0.010 | 0.033 | Pythia-410m FS | M  | 0.063 | 0.024 | 0.033 |
| RoBERTa FT     | M  | 0.086 | 0.004 | 0.025 | Pythia-410m 0s | U  | 0.267 | 0.209 | 0.227 |
| Pythia-1.4B FT | U  | 0.099 | 0.010 | 0.032 | Pythia-410m 0s | M  | 0.083 | 0.107 | 0.122 |
| Pythia-1.4B FT | M  | 0.178 | 0.009 | 0.034 | Pythia-160m FT | U  | 0.000 | 0.001 | 0.001 |
| Pythia-1.4B FS | U  | 0.071 | 0.027 | 0.041 | Pythia-160m FT | M  | 0.007 | 0.003 | 0.002 |
| Pythia-1.4B FS | M  | 0.028 | 0.023 | 0.034 | Pythia-160m FS | U  | 0.015 | 0.020 | 0.024 |
| Pythia-1.4B 0s | U  | 0.534 | 0.061 | 0.092 | Pythia-160m FS | M  | 0.026 | 0.013 | 0.016 |
| Pythia-1.4B 0s | M  | 0.189 | 0.011 | 0.027 | Pythia-160m 0s | U  | 0.473 | 0.206 | 0.227 |
| Pythia-410m FT | U  | 0.043 | 0.001 | 0.005 | Pythia-160m 0s | M  | 0.253 | 0.066 | 0.086 |

## D Full Faithfulness Scores by Profession

Mean faithfulness (probability drop after erasing top-5 attributed tokens) for all 20 professions across both models and masking conditions. U = unmasked; M = masked. Macro-average (equal weight per profession) reported in the *Overall* row.

| Profession        | DistilBERT |       | RoBERTa |       |
|-------------------|------------|-------|---------|-------|
|                   | U          | M     | U       | M     |
| Accountant        | 0.399      | 0.422 | 0.259   | 0.268 |
| Architect         | 0.370      | 0.433 | 0.283   | 0.268 |
| Attorney          | 0.162      | 0.221 | 0.175   | 0.158 |
| Comedian          | 0.352      | 0.296 | 0.240   | 0.292 |
| Composer          | 0.373      | 0.467 | 0.228   | 0.329 |
| Dentist           | 0.353      | 0.390 | 0.271   | 0.228 |
| Dietitian         | 0.620      | 0.743 | 0.548   | 0.372 |
| Filmmaker         | 0.277      | 0.368 | 0.190   | 0.188 |
| Journalist        | 0.268      | 0.275 | 0.281   | 0.318 |
| Model             | 0.294      | 0.153 | 0.100   | 0.192 |
| Nurse             | 0.395      | 0.518 | 0.203   | 0.197 |
| Painter           | 0.473      | 0.418 | 0.339   | 0.342 |
| Photographer      | 0.385      | 0.469 | 0.283   | 0.346 |
| Physician         | 0.233      | 0.191 | 0.235   | 0.200 |
| Poet              | 0.315      | 0.368 | 0.124   | 0.150 |
| Professor         | 0.160      | 0.214 | 0.162   | 0.182 |
| Psychologist      | 0.356      | 0.378 | 0.232   | 0.252 |
| Software Engineer | 0.171      | 0.204 | 0.193   | 0.199 |
| Surgeon           | 0.255      | 0.411 | 0.249   | 0.221 |
| Teacher           | 0.383      | 0.342 | 0.269   | 0.276 |
| <i>Overall</i>    | 0.330      | 0.364 | 0.243   | 0.249 |

## E Gender-Conditioned Faithfulness by Profession

Mean faithfulness scores split by gender for professions with at least 10 examples per gender group, shown for all four conditions (two models  $\times$  masked/unmasked).  $n$  = number of examples. Dietitian is excluded from masked rows (male  $n=3$  in DistilBERT masked;  $n<10$  in all masked conditions).

| Profession                   | Female       |     | Male         |     |
|------------------------------|--------------|-----|--------------|-----|
|                              | Faith.       | n   | Faith.       | n   |
| <i>DistilBERT — unmasked</i> |              |     |              |     |
| Accountant                   | 0.373        | 10  | 0.408        | 31  |
| Architect                    | 0.490        | 17  | 0.326        | 47  |
| Attorney                     | 0.161        | 110 | 0.162        | 160 |
| Dentist                      | 0.350        | 31  | 0.354        | 85  |
| Dietitian                    | 0.656        | 27  | 0.288        | 3   |
| Filmmaker                    | 0.263        | 17  | 0.284        | 31  |
| Journalist                   | 0.238        | 80  | 0.301        | 74  |
| Nurse                        | 0.354        | 132 | <b>0.756</b> | 15  |
| Painter                      | 0.492        | 31  | 0.452        | 28  |
| Photographer                 | 0.432        | 76  | 0.357        | 126 |
| Physician                    | 0.198        | 171 | 0.270        | 157 |
| <b>Poet</b>                  | <b>0.200</b> | 29  | <b>0.434</b> | 28  |
| Professor                    | 0.158        | 413 | 0.162        | 519 |
| Psychologist                 | 0.370        | 79  | 0.336        | 55  |
| Surgeon                      | 0.290        | 15  | 0.249        | 84  |
| Teacher                      | 0.338        | 76  | 0.446        | 55  |
| <i>DistilBERT — masked</i>   |              |     |              |     |
| Accountant                   | 0.413        | 11  | 0.425        | 28  |
| Architect                    | 0.521        | 14  | 0.406        | 47  |
| Attorney                     | 0.223        | 115 | 0.220        | 161 |
| Dentist                      | 0.348        | 31  | 0.406        | 83  |
| Filmmaker                    | 0.310        | 17  | 0.397        | 34  |
| Journalist                   | 0.311        | 79  | 0.235        | 72  |
| Nurse                        | 0.494        | 126 | <b>0.660</b> | 22  |
| Painter                      | 0.476        | 26  | 0.368        | 30  |
| Photographer                 | 0.509        | 76  | 0.444        | 124 |
| Physician                    | 0.183        | 173 | 0.201        | 150 |
| Poet                         | 0.254        | 28  | 0.491        | 26  |
| Professor                    | 0.197        | 422 | 0.227        | 544 |
| Psychologist                 | 0.391        | 69  | 0.359        | 47  |
| Surgeon                      | 0.397        | 19  | 0.414        | 82  |
| Teacher                      | 0.308        | 75  | 0.395        | 48  |
| <i>RoBERTa — unmasked</i>    |              |     |              |     |
| Accountant                   | 0.408        | 12  | 0.197        | 29  |
| Architect                    | 0.381        | 17  | 0.252        | 52  |
| Attorney                     | 0.192        | 108 | 0.164        | 156 |
| Dentist                      | 0.289        | 31  | 0.265        | 84  |
| Dietitian                    | 0.549        | 28  | 0.533        | 2   |
| Filmmaker                    | 0.088        | 20  | 0.244        | 37  |
| Journalist                   | 0.259        | 84  | 0.309        | 68  |
| Nurse                        | 0.163        | 130 | <b>0.533</b> | 16  |
| Painter                      | 0.310        | 28  | 0.367        | 29  |
| Photographer                 | 0.291        | 72  | 0.278        | 131 |
| Physician                    | 0.280        | 174 | 0.184        | 154 |
| Poet                         | 0.139        | 28  | 0.108        | 28  |
| Professor                    | 0.155        | 411 | 0.168        | 533 |
| Psychologist                 | 0.234        | 77  | 0.227        | 52  |
| Surgeon                      | 0.384        | 15  | 0.226        | 86  |
| Teacher                      | 0.249        | 85  | 0.307        | 46  |
| <i>RoBERTa — masked</i>      |              |     |              |     |

(continued on next page)



| Profession   | Female |     | Male         |     |
|--------------|--------|-----|--------------|-----|
|              | Faith. | n   | Faith.       | n   |
| Accountant   | 0.382  | 14  | 0.215        | 30  |
| Architect    | 0.405  | 18  | 0.221        | 53  |
| Attorney     | 0.190  | 109 | 0.134        | 153 |
| Dentist      | 0.241  | 31  | 0.223        | 85  |
| Filmmaker    | 0.177  | 19  | 0.194        | 36  |
| Journalist   | 0.313  | 77  | 0.324        | 74  |
| Nurse        | 0.172  | 129 | <b>0.374</b> | 18  |
| Painter      | 0.363  | 28  | 0.324        | 32  |
| Photographer | 0.418  | 74  | 0.302        | 124 |
| Physician    | 0.218  | 170 | 0.181        | 154 |
| Poet         | 0.142  | 28  | 0.159        | 27  |
| Professor    | 0.172  | 410 | 0.191        | 525 |
| Psychologist | 0.274  | 80  | 0.221        | 55  |
| Surgeon      | 0.247  | 20  | 0.214        | 83  |
| Teacher      | 0.225  | 79  | 0.354        | 52  |

Professions where only one gender reaches  $n \geq 10$  (comedian|M, composer|M, model|F, software engineer|M) are omitted; their overall faithfulness appears in Appendix D. Dietitian|M ( $n=3$  in DU;  $n=2$  in RU;  $n < 10$  in masked conditions) is included for reference but should be interpreted with caution.

The nurse|M asymmetry remains the largest gender-conditioned faithfulness gap in both unmasked conditions: DistilBERT 0.756 vs. 0.354, RoBERTa 0.533 vs. 0.163. A second notable asymmetry appears for *poet* under DistilBERT unmasked (0.434 vs. 0.200), suggesting that male poet predictions are similarly concentrated on a small decisive feature set—though it does not persist under masking (0.491 vs. 0.254) or under RoBERTa (0.108 vs. 0.139). Under masking, the nurse gap narrows substantially (DistilBERT: 0.660 vs. 0.494; RoBERTa: 0.374 vs. 0.172) but does not disappear, consistent with implicit proxy signals persisting after explicit gender-token removal.

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